L04: Positivity, Trimming, Target Populations, and Partial Identification

S&DS 6165: Topics in Causal Inference

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Lecture outline

- Begin with randomized designs that make some treatment comparisons rare or impossible.
- Separate positivity, population overlap, and overlap in one realized experiment.
- Compare clipping with two explicit target changes: trimming and weighting toward equipoise.
- When the original target must remain, replace an unsupported point claim with bounds justified by boundedness, monotonicity, or smoothness.

1 A randomized experiment can still lack a comparison

Consider a randomized trial conducted in three prespecified eligibility strata. The investigator uses different treatment probabilities because the treatment is increasingly difficult to offer:

Stratum	Number of units	Treatment probability	Expected treated units
Standard eligibility	80	0.50	40
Elevated risk	15	0.05	0.75
Logistical exclusion	5	0	0

The first stratum supplies an ordinary randomized comparison. Treatment remains possible in the elevated-risk stratum, but it is rare: the probability that none of its 15 units is treated is $0.95^{15} \approx 0.46$. A logistical rule makes treatment impossible in the final stratum under the experiment as conducted.

Suppose the scientific target is the finite-population average treatment effect over all 100 trial units. Two different problems arise.

First, the treatment effects of the five logistically excluded units enter the target, but their treated outcomes can never be exposed by this design. This is an **identification** problem. Repeating the same experiment indefinitely would not reveal the missing comparison.

Second, a treated elevated-risk unit can appear, but there may be only one—or none—in the realized experiment. An inverse-probability estimator asks any one such unit to represent about $1/0.05 = 20$ similar units. This is a **precision and finite-sample stability** problem. The comparison exists in the design, but the realized experiment may contain little information about it.

The distinction between zero and merely small probability is the central distinction of this lecture. It also determines which remedies can be honest: no amount of weight regularization turns a structurally impossible assignment into randomized evidence.

2 Positivity in a finite randomized design

Return to the finite population of N units used in L01–L03. Potential outcomes $Y_i(1)$ and $Y_i(0)$, covariates X_i, and the target

$$\tau = \frac{1}{N}\sum_{i=1}^{N}\{Y_i(1) - Y_i(0)\}$$

are fixed. The assignment vector $\mathbf{Z}$ is random under a known design. For each unit and treatment state, define the marginal design probability

$$\pi_i(z) = \mathbb{P}_Z(Z_i = z), \qquad z \in \{0, 1\}.$$

For a binary treatment, $\pi_i(1) = \pi_i$ and $\pi_i(0) = 1 - \pi_i$. Design positivity for the finite-population ATE means

$$\pi_i(0) > 0 \quad \text{and} \quad \pi_i(1) > 0 \qquad \text{for every } i. \tag{1}$$

The condition is relative to the estimand. If unit i contributes both potential outcomes to τ, the design must give both outcomes a chance to be observed. If the target deliberately excludes unit i, its assignment probabilities need not identify that restricted target.

This is a condition on marginal exposure probabilities, not on every complete assignment vector. Under complete randomization, most of the 2^N binary assignment vectors have probability zero because the treated count is fixed. Nevertheless, every unit can appear in either arm, so Equation 1 holds whenever both arm sizes are positive.

> **i** Positivity and overlap are related, but not identical
>
> **Positivity** is a property of a treatment or exposure mechanism relative to a target: every state needed for the target has positive probability. In a randomized experiment, this is known from the design.
> **Population overlap** usually describes the same support requirement through covariates. If assignment depends on X, the treated and control populations overlap at x when both treatment probabilities are positive there. In an observational study this becomes $0 < e(x) < 1$, where $e(x) = \mathbb{P}(Z = 1 \mid X = x)$.
> **Realized covariate overlap** describes the sample actually obtained. Even when the design has positivity, random assignment can leave a small stratum with only treated or only control units. That realization may make estimation unstable, but it does not retroactively change the design probabilities. Conversely, seeing nearby treated and control units in a sample does not prove that an observational assignment mechanism is unconfounded.

In practice, "weak positivity" and "weak overlap" are often used for probabilities close to zero or one. Strictly speaking, positivity is binary: the probability is positive or it is not. The adjective *weak* signals that a formally positive design may contain so little information in some regions that its estimators behave poorly.

A stronger quantitative condition is often called **uniform** or **strong positivity**: for some fixed $\varepsilon > 0$,

$$\varepsilon \le \pi_i \le 1 - \varepsilon \qquad \text{for every target unit } i.$$

Ordinary positivity permits probabilities arbitrarily close to zero or one; strong positivity rules them out.

3 What the inverse probability is doing

The basic identity is already present in the Horvitz–Thompson estimator:

$$\mathbb{E}_Z\left\{\frac{\mathbf{1}\{Z_i = z\}Y_i^{\text{obs}}}{\pi_i(z)}\right\} = Y_i(z), \tag{2}$$

provided $\pi_i(z) > 0$. The indicator reveals $Y_i(z)$ only with probability $\pi_i(z)$; division by that known probability restores one full unit of that potential outcome in design expectation.

L03 used the same identity to correct prediction errors rather than outcome levels. Let $g_z(X_i)$ be a prediction fixed before assignment and define the fixed residual

$$r_i(z) = Y_i(z) - g_z(X_i).$$

Under independent Bernoulli assignment with unit-specific probabilities π_i, consider

$$\hat{\tau}(g) = \frac{1}{N}\sum_{i=1}^{N}\left[g_1(X_i) - g_0(X_i) + \frac{Z_i r_i(1)}{\pi_i} - \frac{(1 - Z_i)r_i(0)}{1 - \pi_i}\right]. \tag{3}$$

The prediction supplies a treatment contrast for every unit. Randomization then corrects its errors. When $0 < \pi_i < 1$,

$$\mathbb{E}_Z\left\{\frac{Z_i r_i(1)}{\pi_i}\right\} = r_i(1), \qquad \mathbb{E}_Z\left\{\frac{(1 - Z_i)r_i(0)}{1 - \pi_i}\right\} = r_i(0),$$

so $\mathbb{E}_Z\{\hat{\tau}(g)\} = \tau$ for any fixed predictions. The inverse probability is not a generic penalty for unlikely observations. It undoes the probability with which the design exposes the corresponding residual.

If $\pi_i = 0$, the treated identity and residual correction do not become "infinitely large." They cease to be defined, because $Z_i Y_i(1)$ is always zero and the design never reveals $Y_i(1)$. A model can still supply $g_1(X_i)$, but randomization cannot check or correct that model at unit i. The resulting claim is an extrapolation claim rather than a design-unbiased one.

3.1 Weak probabilities enter the variance

The same representation shows what happens before probability reaches zero. Subtracting the design expectation of the ith summand in Equation 3 gives

$$(Z_i - \pi_i)\left\{\frac{r_i(1)}{\pi_i} + \frac{r_i(0)}{1-\pi_i}\right\}. \tag{4}$$

Under independent Bernoulli assignment, contributions from different units are independent and $\mathrm{Var}_Z(Z_i) = \pi_i(1-\pi_i)$. Therefore

$$\mathrm{Var}_Z\{\hat{\tau}(g)\} = \frac{1}{N^2}\sum_{i=1}^{N} \pi_i(1-\pi_i)\left\{\frac{r_i(1)}{\pi_i} + \frac{r_i(0)}{1-\pi_i}\right\}^2. \tag{5}$$

The formula makes two mechanisms visible.

1. A small π_i magnifies the treated residual $r_i(1)$; a value near one magnifies the control residual $r_i(0)$.
2. Better fixed predictions can reduce the residuals, but cannot change which assignments the design permits.

The formula is exact for independent Bernoulli assignment. Under complete randomization, rerandomization, or another dependent design, cross-unit covariances also enter. The support conclusion is unchanged: weighting can correct unequal positive probabilities, but it cannot manufacture a treatment state with probability zero.

4 Four responses, four different claims

Once support is inadequate, there is no method that simultaneously preserves the original target, relies only on the original randomized evidence, retains a point conclusion, and enjoys stable precision. Something must change.

Response	What changes?	What can be claimed?
Clip or cap inverse weights	The estimator	A more stable but generally biased point estimate
Trim the population	The target units	A point estimate for an explicitly restricted population
Weight toward equipoise	The target weights	A point estimate for a smooth, design-supported population
Bound unsupported outcomes	The form of the conclusion and the assumptions	An identified set for the original target

A fifth response is often best: redesign the experiment, recruit different units, or collect more data in the poorly represented strata. Statistical adjustments are not automatically preferable to fixing the source of the information problem.

5 Clipping: regularize an estimator

Begin with the unaugmented Horvitz–Thompson estimator,

$$\hat{\tau}_{\mathrm{HT}} = \frac{1}{N}\sum_{i=1}^{N}\left\{\frac{Z_i Y_i^{\mathrm{obs}}}{\pi_i} - \frac{(1-Z_i)Y_i^{\mathrm{obs}}}{1-\pi_i}\right\}.$$

Suppose all probabilities are positive but some inverse probabilities are uncomfortably large. Choose a cap $C > 1$ and replace each inverse probability by

$$c_{i1} = \min\left(\frac{1}{\pi_i}, C\right), \qquad c_{i0} = \min\left(\frac{1}{1-\pi_i}, C\right). \tag{6}$$

This operation is called weight **clipping**, **truncation**, or **Winsorization**. The clipped estimator is

$$\hat{\tau}_C = \frac{1}{N}\sum_{i=1}^{N}\{Z_i c_{i1} Y_i^{\mathrm{obs}} - (1-Z_i)c_{i0}Y_i^{\mathrm{obs}}\}. \tag{7}$$

Clipping limits how much any rare assignment can contribute. It also breaks the cancellation that made Horvitz–Thompson unbiased. Taking the design expectation gives

$$\mathbb{E}_Z(\hat{\tau}_C) = \frac{1}{N}\sum_{i=1}^{N}\left[\min(1, C\pi_i)Y_i(1) - \min\{1, C(1-\pi_i)\}Y_i(0)\right]. \tag{8}$$

Whenever $\pi_i < 1/C$, the treated potential outcome is attenuated by $C\pi_i$; whenever $1-\pi_i < 1/C$, the control potential outcome is attenuated by $C(1-\pi_i)$. These arm-specific factors generally do not define one common target population. Thus clipping is not automatically unbiased for a neatly reweighted causal estimand.

For example, if $\pi_i = 0.02$ and $C = 10$, the treated contribution appears in expectation with multiplier $C\pi_i = 0.2$, not one. The cap removes 80 percent of that unit's treated mean contribution in exchange for limiting its realized weight to ten.

Normalizing the clipped treated and control weights to sum to one does not restore the original ATE. It makes each arm represent its own tilted population, and the two tilts need not agree. The practical gain is variance control; the price is design bias and increased dependence on outcome patterns in the clipped region.

Clipping can still be useful as an explicit regularization or sensitivity analysis. A credible report varies C and shows how the estimate, the largest weight, and an effective-sample-size diagnostic change. It should not present the clipped estimate as though the original randomization identity still held. Cole and Hernán and Lee, Lessler, and Stuart discuss this practical use of truncated inverse weights in observational analyses (2008; 2011).

Clipping also cannot repair a structural zero. If $\pi_i = 0$, writing down a finite substitute for $1/\pi_i$ does not create a treated observation for unit i.

6 Trimming: change the target explicitly

A more transparent response is to name the population the design can reasonably represent. Let $A \subseteq \{1, \dots, N\}$ be a set of units defined using design probabilities and pretreatment information, before outcomes are examined, and require $0 < \pi_i < 1$ for every $i \in A$. Write $N_A = |A|$. The finite-population effect in this restricted target is

$$\tau_A = \frac{1}{N_A} \sum_{i \in A} \{Y_i(1) - Y_i(0)\}. \tag{9}$$

For fixed predictions $g_z(X_i)$, a design-unbiased estimator is

$$\hat{\tau}_A(g) = \frac{1}{N_A} \sum_{i \in A} \left[g_1(X_i) - g_0(X_i) + \frac{Z_i r_i(1)}{\pi_i} - \frac{(1 - Z_i) r_i(0)}{1 - \pi_i} \right]. \tag{10}$$

This estimator does not pretend to recover the effects of excluded units. It estimates a different, explicitly stated estimand. If the five logistically excluded units in the opening example are excluded, positivity may hold for the remaining 95 units. Excluding the elevated-risk stratum as well may produce a more stable estimate for 80 standard-eligibility units, but it changes the question again.

Under Bernoulli assignment, the exact residual variance of Equation 10 is

$$\mathrm{Var}_Z\{\hat{\tau}_A(g)\} = \frac{1}{N_A^2} \sum_{i \in A} v_i, \tag{11}$$

where

$$v_i = \pi_i(1 - \pi_i) \left\{ \frac{r_i(1)}{\pi_i} + \frac{r_i(0)}{1 - \pi_i} \right\}^2. \tag{12}$$

Removing a unit with a large v_i reduces the numerator but also reduces N_A, which increases the normalization applied to every retained unit. Trimming is therefore a **precision–relevance tradeoff**, not a rule to delete every difficult observation.

There is an important practical obstacle: v_i contains both potential outcomes and is not known before assignment. A feasible rule must replace it with a prespecified proxy or an estimate based on pretreatment covariates and appropriately separated outcome information. Once the same data choose the target and estimate its effect, inference must account for that adaptivity.

6.1 The superpopulation counterpart

The modern trimming literature is usually written for an i.i.d. superpopulation. Let X be pretreatment covariates,

$$e(x) = \mathbb{P}(Z = 1 \mid X = x), \qquad m_z(x) = \mathbb{E}\{Y(z) \mid X = x\},$$

and let

$$\sigma_z^2(x) = \mathrm{Var}\{Y(z) \mid X = x\}.$$

In an observational study, interpreting these quantities causally also requires consistency and conditional exchangeability; overlap alone is not enough. In a randomized design whose probabilities depend on X, the assignment probabilities are known and exchangeability follows from the design.

For a covariate set $\mathcal{A}$ with positive population probability, the target is

$$\tau_{\mathcal{A}} = \mathbb{E}\{Y(1) - Y(0) \mid X \in \mathcal{A}\}. \tag{13}$$

Crump, Hotz, Imbens, and Mitnik study the choice of such target populations when overlap is limited (2009). A central conditional residual-variance score is

$$k(x) = \frac{\sigma_1^2(x)}{e(x)} + \frac{\sigma_0^2(x)}{1 - e(x)}. \tag{14}$$

For a retained covariate set $\mathcal{A}$, the corresponding residual criterion is

$$V(\mathcal{A}) = \frac{\mathbb{E}[\mathbf{1}\{X \in \mathcal{A}\}k(X)]}{\mathbb{P}(X \in \mathcal{A})^2}. \tag{15}$$

Its minimizer has a threshold form: retain covariate values with $k(x) \leq \gamma$ for an appropriate γ. Under equal homoskedastic conditional variances,

$$k(x) = \frac{\sigma^2}{e(x)\{1 - e(x)\}},$$

so this rule reduces to a symmetric propensity-score region such as $\alpha \leq e(x) \leq 1 - \alpha$. A conventional cutoff like 0.1 is not a universal optimum; the appropriate set depends on the probability and outcome structure.

Khan and Ugander extend this logic to heteroscedastic outcomes and develop doubly robust, sample-adaptive trimming procedures (2025). Trimming only on $e(x)$ can discard a rare but highly predictable region or retain a noisy region near one-half. The more refined score recognizes that weak overlap and residual uncertainty jointly determine precision.

The criterion in Equation 15 conditions on the covariates and focuses on residual variation. The full superpopulation variance also contains heterogeneity in $m_1(X) - m_0(X)$ within the target. One must specify the inferential regime before calling a trimming rule "optimal."

6.2 A numerical precision–relevance tradeoff

Suppose 90 percent of a source population has $e(X) = 0.5$ and 10 percent has $e(X) = 0.01$. Let both conditional residual variances equal one. The full-population residual criterion has numerator

$$0.9(4) + 0.1(101.01) \approx 13.70.$$

If the rare-treatment region is removed, the numerator is $0.9(4)$ and the retained mass is 0.9, giving

$$V(\mathcal{A}) = \frac{0.9(4)}{0.9^2} \approx 4.44.$$

The precision gain is large. Yet the new estimand omits the ten percent of the population for whom treatment was rare. If those units are the policy population of greatest concern, a precise answer about the other 90 percent does not answer the original question.

7 Equipoise: change the target smoothly

Hard trimming draws a boundary: a unit just inside the cutoff belongs fully to the target, while a nearly identical unit just outside does not belong at all. A smoother alternative is to give greater target weight to units for whom both treatment states were genuinely available.

In the finite randomized population, let $h_i \geq 0$ be a prespecified target weight and define

$$\tau_h = \frac{\sum_{i=1}^{N} h_i\{Y_i(1) - Y_i(0)\}}{\sum_{i=1}^{N} h_i}. \tag{16}$$

Choose

$$h_i = \pi_i(1 - \pi_i). \tag{17}$$

This weight is largest at $\pi_i = 1/2$, shrinks as the design makes one arm rare, and equals zero when one arm is impossible. A Horvitz–Thompson estimator for this target is

$$\hat{\tau}_h = \frac{1}{\sum_i h_i} \sum_{i:\, 0<\pi_i<1} h_i \left\{ \frac{Z_i Y_i^{\text{obs}}}{\pi_i} - \frac{(1 - Z_i)Y_i^{\text{obs}}}{1 - \pi_i} \right\}. \tag{18}$$

For units with positive probabilities, the observed arm weights simplify:

$$\frac{h_i}{\pi_i} = 1 - \pi_i \quad \text{when } Z_i = 1, \qquad \frac{h_i}{1 - \pi_i} = \pi_i \quad \text{when } Z_i = 0. \tag{19}$$

Units with $\pi_i \in \{0, 1\}$ have $h_i = 0$ and are omitted from the displayed sum so that no 0/0 expression is formed. The weights on supported units are bounded. More importantly, the stabilization is

not hidden clipping: both arms represent the same explicitly defined target τ_h. A structurally unsupported unit receives zero target weight rather than a fabricated inverse weight.

This target is often described as the **population in equipoise**: units for whom either treatment is plausible under the assignment mechanism. It can be more principled than an arbitrary hard cutoff because membership changes smoothly and the estimand emphasizes the part of the population that supports the comparison.

The interpretation requires care in a designed experiment. The investigator chooses π_i, so changing randomization probabilities changes the equipoise-weighted estimand. Under ordinary equal-probability randomization, every h_i is the same and $\tau_h = \tau$. The target becomes distinctive only when treatment probabilities vary across units or strata.

In a superpopulation, a nonnegative tilting function $h(x)$ defines

$$\tau_h = \frac{\mathbb{E}[h(X)\{Y(1) - Y(0)\}]}{\mathbb{E}\{h(X)\}}. \tag{20}$$

Li, Morgan, and Zaslavsky choose $h(x) = e(x)\{1 - e(x)\}$, yielding treated weights $1 - e(x)$ and control weights $e(x)$ (2018). The target density is proportional to $e(x)\{1 - e(x)\}f_X(x)$, not to $e(x)\{1-e(x)\}$ alone. It emphasizes covariate regions with treatment uncertainty while still reflecting how common those regions are in the source population.

7.1 What changes in an observational study?

The weighting algebra looks similar, but the source of validity is different. In a randomized experiment, assignment probabilities are selected and known, and design expectations hold without an outcome model. In an observational study:

- $e(x)$ must be estimated;
- conditional exchangeability is a substantive assumption, not a consequence of design;
- unmeasured confounding can remain even when propensity distributions look well overlapped; and
- high-dimensional positivity is difficult to assess from one-dimensional propensity histograms.

Thus good observed overlap is useful for estimation but does not itself make an observational comparison causal. D'Amour et al. explain why overlap becomes especially demanding with many covariates (2021).

8 When the target cannot be changed

Clipping regularizes a point estimator. Trimming and equipoise weighting name a different target population. Sometimes the original target is scientifically nonnegotiable. The five logistically excluded units in the opening example may be precisely the people a policy decision must address.

If the design never exposes one of their potential outcomes, a point effect for the full target requires extrapolation. **Partial identification** offers a different response: retain the original target, state

additional restrictions that are scientifically defensible, and report every effect value compatible with the design, the observed-data law, and those restrictions.

Definition 8.1 (Identified set)**.** For an estimand θ and an assumption class $\mathcal{M}$, the **identified set** is the set of values of θ attained by full-data distributions—or finite science tables in a design-based analysis—that produce the same observed-data law and satisfy $\mathcal{M}$.

A bound is **sharp** when every value inside it is attainable, or can be approached arbitrarily closely, by some admissible full-data world. An identified set is a population object. Its width can remain positive even with infinite repeated data; sampling uncertainty about its estimated endpoints is a separate issue.

9 Manski's accounting: bound what the design cannot expose

Assume the outcomes are known to lie in

$$Y_i(z) \in [\ell, u] \qquad \text{for every } i, z. \tag{21}$$

For arm z, separate units into

$$\mathcal{J}_z^+ = \{i : \pi_i(z) > 0\}, \qquad \mathcal{J}_z^0 = \{i : \pi_i(z) = 0\}.$$

The design can expose $Y_i(z)$ for every $i \in \mathcal{J}_z^+$ in some assignment. It can never expose $Y_i(z)$ for $i \in \mathcal{J}_z^0$. Two finite science tables that agree on all supported potential outcomes but differ on the unsupported ones generate the same observed-data distribution under the design.

Let

$$\mu_z = \frac{1}{N} \sum_{i=1}^{N} Y_i(z).$$

Boundedness yields

$$\mu_z^- = \frac{1}{N} \left\{ \sum_{i \in \mathcal{J}_z^+} Y_i(z) + |\mathcal{J}_z^0| \ell \right\}, \qquad \mu_z^+ = \frac{1}{N} \left\{ \sum_{i \in \mathcal{J}_z^+} Y_i(z) + |\mathcal{J}_z^0| u \right\}. \tag{22}$$

Combining the two arm intervals gives the ATE bounds

$$\tau \in \left[\mu_1^- - \mu_0^+, \mu_1^+ - \mu_0^- \right]. \tag{23}$$

The lower effect pairs the smallest permissible treated mean with the largest permissible control mean; the upper effect does the reverse. Under only Equation 21, these bounds are sharp because each unsupported potential outcome can be set independently to an endpoint of its allowable range.

The supported sums in Equation 22 are not all observed in one realization, but they are design-estimable. For example,

$$\hat{\mu}_z^- = \frac{1}{N}\left\{\sum_{i\in\mathcal{J}_z^+}\frac{\mathbf{1}\{Z_i = z\}Y_i^{\text{obs}}}{\pi_i(z)} + |\mathcal{J}_z^0|\ell\right\} \tag{24}$$

is design-unbiased for μ_z^-. Replacing ℓ by u estimates the upper endpoint. This is where identification and inference separate: the population endpoints are determined by assumptions and the design, while a single randomized experiment estimates those endpoints with sampling error.

In the opening example, suppose only the five logistically excluded units lack treated support, control is supported for everyone, and outcomes lie in $[0, 100]$. Their unknown treated outcomes contribute between 0 and $5 \times 100/100 = 5$ points to the population treated mean. Even perfect knowledge about the other 95 units leaves an ATE identified interval of width five. More repetitions improve estimation of the supported contribution but do not narrow this design-imposed gap.

9.1 The familiar observational form

Manski's worst-case logic applies without a randomized design as well (1990). Let $\rho = \mathbb{P}(Z = 1)$ and write

$$m_1^{\text{obs}} = \mathbb{E}(Y^{\text{obs}} \mid Z = 1), \qquad m_0^{\text{obs}} = \mathbb{E}(Y^{\text{obs}} \mid Z = 0).$$

Consistency identifies the treated group's mean of $Y(1)$ and the control group's mean of $Y(0)$, but not the two missing counterfactual means. Under $Y(z) \in [\ell, u]$,

$$\mu_1 \in \left[\rho m_1^{\text{obs}} + (1-\rho)\ell, \rho m_1^{\text{obs}} + (1-\rho)u\right], \tag{25}$$

and

$$\mu_0 \in \left[(1-\rho)m_0^{\text{obs}} + \rho\ell, (1-\rho)m_0^{\text{obs}} + \rho u\right]. \tag{26}$$

The ATE bounds again combine the smallest treated mean with the largest control mean, and vice versa. Their width is not a small-sample defect. More observational data estimate the same endpoints more precisely; they do not identify the missing counterfactual means without additional assumptions.

10 Lee bounds: use randomized selection structure

Bounded outcomes can produce wide intervals. Credible structure can narrow them. Lee's bounds provide a particularly instructive randomized example (2009).

Consider a randomized job-training program. Let $S_i(z) \in \{0, 1\}$ indicate whether unit i's wage would be observed under assignment z—for example, whether the person would be employed. Let $Y_i(z)$ be the wage defined when $S_i(z) = 1$. The observed selection indicator is

$$S_i = Z_i S_i(1) + (1 - Z_i) S_i(0),$$

and a wage is observed only when $S_i = 1$. Although treatment is randomized, comparing wages among employed treated and employed control units need not be a causal comparison: treatment may change who becomes employed.

Assume random assignment and **monotone selection**,

$$S_i(1) \geq S_i(0) \qquad \text{for every } i. \tag{27}$$

Monotonicity rules out a unit who would be employed under control but not under treatment. It leaves three principal strata: always employed, employed only when treated, and never employed. The target is the effect among the always-employed stratum,

$$\tau_{\mathrm{AO}} = \mathbb{E}\{Y(1) - Y(0) \mid S(1) = S(0) = 1\}.$$

Let

$$s_z = \mathbb{P}(S = 1 \mid Z = z),$$

and suppose $0 < s_0 \leq s_1$. Randomization identifies both selection rates. Under monotonicity, every employed control unit is always employed. Among employed treated units, the fraction

$$q = \frac{s_0}{s_1} \tag{28}$$

is always employed; the remaining fraction $1 - q$ is employed only because of treatment. The data identify this mixture proportion but not which treated wages belong to which stratum.

Let $Q_1(v)$ denote the quantile function of wages among employed treated units. The smallest possible treated mean for the always-employed group assigns that group the bottom fraction q of treated wages. The largest assigns it the top fraction q:

$$\mu_{1,L}^{\mathrm{AO}} = \frac{1}{q} \int_0^q Q_1(v)\, dv, \qquad \mu_{1,U}^{\mathrm{AO}} = \frac{1}{q} \int_{1-q}^1 Q_1(v)\, dv. \tag{29}$$

The employed-control mean

$$\mu_0^{\mathrm{AO}} = \mathbb{E}(Y^{\mathrm{obs}} \mid Z = 0, S = 1)$$

identifies the control mean for the always-employed stratum. Therefore

$$\tau_{\mathrm{AO}} \in \left[\mu_{1,L}^{\mathrm{AO}} - \mu_0^{\mathrm{AO}}, \mu_{1,U}^{\mathrm{AO}} - \mu_0^{\mathrm{AO}}\right]. \tag{30}$$

If $s_1 = 0.8$ and $s_0 = 0.6$, then $q = 0.75$. The lower bound removes the top 25 percent of employed treated wages and averages the bottom 75 percent; the upper bound removes the bottom 25 percent and averages the top 75 percent. With ties at a cutoff, fractional trimming may be required.

This outcome trimming has a different purpose from Crump-style covariate trimming. Crump-style trimming changes the target population to improve support and precision. Lee trimming constructs extreme allocations within a latent mixture to bound a fixed principal-stratum effect.

The inequality $s_1 \geq s_0$ in the observed data is necessary for the stated direction of monotonicity, but it does not prove individual-level monotonicity. That substantive assumption is what narrows the identified set. In return, the Lee construction does not require a bounded wage scale.

11 Smoothness: controlled borrowing across support

Boundedness treats a response just outside support as unrelated to a response just inside. A parametric model takes the opposite extreme: after its form and parameters are fixed, it selects one extrapolated response. A smoothness restriction can occupy the middle ground.

Consider a target population with covariates X and a randomized **behavior policy** that chooses arm z with known probability $b_z(x)$. The policy may never assign an arm in some covariate regions. Let

$$\mathcal{O}_z = \{x : b_z(x) > 0\}$$

be the support for arm z, and define the conditional potential-outcome mean

$$m_z(x) = \mathbb{E}\{Y(z) \mid X = x\}.$$

Randomization identifies $m_z(x)$ on $\mathcal{O}_z$, but not outside it. Suppose a new target policy would use arm z outside $\mathcal{O}_z$. Off-policy evaluation then requires information that the behavior design did not directly collect.

Choose a distance $d(x, x')$ on the covariate space and a constant $L_z \geq 0$. Assume

$$|m_z(x) - m_z(x')| \leq L_z d(x, x') \tag{31}$$

for covariate values relevant to the target. This constrains the conditional mean, not every noisy individual outcome. The identified function on $\mathcal{O}_z$ must itself satisfy this restriction; otherwise the assumption class is empty. Every supported point x' implies

$$m_z(x') - L_z d(x, x') \le m_z(x) \le m_z(x') + L_z d(x, x').$$

All such restrictions must hold simultaneously. The tight pointwise envelopes outside support are

$$\underline{m}_z(x) = \sup_{x' \in \mathcal{O}_z} \{m_z(x') - L_z d(x, x')\}, \tag{32}$$

and

$$\overline{m}_z(x) = \inf_{x' \in \mathcal{O}_z} \{m_z(x') + L_z d(x, x')\}. \tag{33}$$

If outcomes are also bounded in $[\ell, u]$, truncate the lower envelope at ℓ and the upper envelope at u. Averaging the completed envelopes under the target policy and target covariate distribution produces bounds on its value. Khan, Saveski, and Ugander develop sharp smoothness-based bounds for off-policy evaluation beyond overlap (2024).

11.1 A one-dimensional example

Let X be uniform on $[0, 1]$. Suppose arm 1 is supported only on $[0, 0.6]$, where

$$m_1(x) = 0.4 + 0.5x.$$

Assume $Y(1) \in [0, 1]$ and that m_1 is 0.5-Lipschitz. At the support boundary, $m_1(0.6) = 0.7$. For $x > 0.6$, the sharp envelopes are

$$\underline{m}_1(x) = 1 - 0.5x, \qquad \overline{m}_1(x) = 0.4 + 0.5x.$$

At $x = 0.9$, smoothness restricts the missing mean to $[0.55, 0.85]$ rather than the boundedness-only interval $[0, 1]$.

Integration makes the gain for the population arm mean concrete. The supported contribution is

$$\int_0^{0.6} (0.4 + 0.5x)\, dx = 0.33.$$

The lower and upper unsupported contributions are

$$\int_{0.6}^1 (1 - 0.5x)\, dx = 0.24, \qquad \int_{0.6}^1 (0.4 + 0.5x)\, dx = 0.32.$$

Thus

$$\mathbb{E}\{Y(1)\} \in [0.57, 0.65].$$

Boundedness alone gives $[0.33, 0.73]$. Smoothness narrows the answer without selecting one extrapolated curve.

The gain is purchased with substantive choices.

- The metric d declares which covariate changes make units similar. Rescaling a coordinate changes the restriction.
- The constant L_z limits how quickly the mean may change. The unsupported data cannot verify that limit where it matters most.
- In high dimensions, supported points may be far away under any credible metric, so the envelopes remain wide and their estimation can be slow.

A persuasive analysis therefore reports bounds over defensible choices of d and L_z rather than treating them as innocuous tuning parameters.

12 Identification is not inference

An identified interval $[L, U]$ states what the population design or observed-data law and assumptions imply. In a realized sample, both endpoints must be estimated. The plug-in interval $[\widehat{L}, \widehat{U}]$ is not automatically a 95 percent confidence interval.

Two inferential goals are common:

1. construct a confidence region that covers the **entire identified set** $[L, U]$ with a stated probability;
2. construct a confidence interval that covers the unknown scalar effect, knowing only that it lies somewhere in $[L, U]$.

The first goal is stronger and generally produces a wider interval. The second can use the fact that one scalar cannot simultaneously lie near both endpoints. Either way, sampling uncertainty in the estimated endpoints—and their dependence—must be handled. Imbens and Manski give foundational results for this distinction (2004).

When both endpoint estimators are regular and asymptotically normal, a Bonferroni construction can conservatively expand the lower endpoint downward and the upper endpoint upward by $z_{1-\alpha/2}$ one-sided standard errors. Their joint covariance can yield a less conservative critical value. This familiar calculation is not automatic for endpoints produced by max–min operations, moment selection, or nonregular nuisance estimates. For smoothness bounds in particular, nuisance estimation and envelope optimization can converge more slowly than $n^{-1/2}$, so ordinary Wald reasoning needs particular care (Khan, Saveski, and Ugander 2024).

Keep three uncertainties separate:

- **Identification uncertainty:** $U - L$ remains positive even with infinite data under the maintained assumptions.
- **Sampling uncertainty:** L and U are estimated from one finite experiment or sample.
- **Sensitivity uncertainty:** the set changes when the outcome bounds, monotonicity direction, covariate metric, or smoothness constant changes.

Calling all three "the confidence interval" obscures which uncertainty comes from random sampling and which comes from scientific assumptions.

13 What to carry forward

Positivity is not a ritual condition attached to a weighting formula. It says that the design can expose every comparison required by the target estimand. Overlap describes how that support appears across covariate regions; realized overlap can still be poor by chance even under a positive design.

When probabilities are small but positive, inverse weighting remains centered on the design-based target, but residual noise is amplified. Clipping can reduce that amplification by accepting design bias. Trimming and equipoise weighting can yield stable, design-supported point estimates by naming a different target. Those target changes may be scientifically appropriate, but they should never be hidden as mere data cleaning.

When the original target must remain and a comparison has zero probability, partial identification changes the form of the answer. Boundedness, monotonicity, or smoothness can rule out some missing science tables without pretending that the data selected one of them. Stronger assumptions narrow the set; a better design can remove the gap.

14 Advanced material

14.1 Why a variance-optimal trimming rule has a threshold form

The threshold structure in Equation 15 follows from a local calculation. Write

$$q = \mathbb{P}(X \in \mathcal{A}), \qquad K = \mathbb{E}[\mathbf{1}\{X \in \mathcal{A}\}k(X)], \qquad V(\mathcal{A}) = \frac{K}{q^2}.$$

Consider adding an infinitesimal excluded covariate cell of mass ε and difficulty score k_0. The new criterion is

$$V_\varepsilon = \frac{K + \varepsilon k_0}{(q+\varepsilon)^2}.$$

Differentiating at zero gives

$$\left.\frac{dV_\varepsilon}{d\varepsilon}\right|_{\varepsilon=0} = \frac{k_0 q - 2K}{q^3}. \tag{34}$$

Adding the cell improves precision exactly when

$$k_0 < \frac{2K}{q} = 2\mathbb{E}\{k(X) \mid X \in \mathcal{A}\}.$$

At an optimum, retained cells therefore lie below a common threshold and excluded cells lie above it, apart from ties. The threshold satisfies

$$\gamma = 2\mathbb{E}\{k(X) \mid k(X) \leq \gamma\}.$$

This argument concerns the residual, covariate-conditional criterion. For a superpopulation ATE, the full influence-function variance also contains

$$\frac{\mathbb{E}[\mathbf{1}\{X \in \mathcal{A}\}\{m_1(X) - m_0(X) - \tau_{\mathcal{A}}\}^2]}{\mathbb{P}(X \in \mathcal{A})^2}.$$

Treatment-effect heterogeneity can therefore alter which population is variance-optimal.

14.2 Adaptive trimming changes the inferential problem

Suppose an estimated score $\hat{k}$ and threshold $\hat{\gamma}$ define

$$\widehat{\mathcal{A}} = \{x : \hat{k}(x) \leq \hat{\gamma}\}.$$

For sampled covariates $X_1, \ldots, X_n$, the realized target is

$$\tau_{\widehat{\mathcal{A}}}^{\text{sample}} = \frac{\sum_{i=1}^n \mathbf{1}\{X_i \in \widehat{\mathcal{A}}\}\{m_1(X_i) - m_0(X_i)\}}{\sum_{i=1}^n \mathbf{1}\{X_i \in \widehat{\mathcal{A}}\}}.$$

This differs from the effect for a fixed population rule $\overline{\mathcal{A}}$ defined by population quantities. The two are close only under conditions on boundary estimation and the probability mass near that boundary. Khan and Ugander develop cross-fitted, doubly robust inference for the realized sample-adaptive target, including simultaneous intervals over prespecified trimming procedures (2025). That result does not automatically become inference for a fixed limiting target. Data-adaptive target selection is part of the inferential question, not preprocessing to forget.

14.3 Extrapolation as a bias–bias–variance tradeoff

Partial identification is not the only way to make extrapolation assumptions visible. Many regression and weighting estimators predict at a target covariate value x^* through

$$\widehat{m}(x^*) = \sum_{i=1}^n w_i Y_i.$$

If the weights are nonnegative and sum to one, the weighted covariates remain in the convex hull of the observed covariates. When x^* lies outside that hull, nonnegative weights cannot reproduce it exactly. Negative weights can reduce covariate imbalance, but then prediction depends on the outcome model continuing beyond the observed convex support and the weights may amplify noise.

Arbour, Parikh, Niknam, Stuart, Rudolph, and Feller formalize this as a bias–bias–variance tradeoff (2026). To isolate the geometry, one affine-normalized schematic variant of their regularized objective is

$$\min_{w} \left\| \sum_{i=1}^{n} w_i X_i - x^* \right\|_2^2 + \lambda \|w\|_2^2 + \gamma \sum_{i=1}^{n} (-w_i)_+ \|X_i\|^\alpha \quad \text{subject to} \quad \sum_{i=1}^{n} w_i = 1. \tag{35}$$

The first term measures imbalance: how far the weighted covariates remain from the target. The second is a variance proxy that penalizes dispersed weights. The third penalizes negative-weight extrapolation, where $(-w_i)_+ = \max(-w_i, 0)$. The parameters λ and γ regulate two different risks. The affine constraint shown here makes the convex-hull interpretation explicit; equivalent normalization can instead be built into the features, and the precise regularization depends on the outcome-model class and feature scaling.

This framework is not partial identification: it produces a regularized point estimate rather than a sharp set. Its connection to this lecture is conceptual. Poor support forces the analyst to negotiate among target fidelity, model extrapolation, and precision. A sensitivity analysis over γ can show how much the answer depends on permitting negative influence rather than concealing that dependence in a single fitted model.

15 Further reading

- Crump et al. (2009) formulate limited overlap as a target-population and optimal-trimming problem. Yang and Ding (2018) study inference when trimming uses an estimated propensity score.
- Khan and Ugander (2025) extend trimming to heteroscedastic outcomes and sample-adaptive, doubly robust inference. Li, Morgan, and Zaslavsky (2018) develop balancing weights and the overlap-weighted population in equipoise.
- Manski (1990) gives foundational nonparametric treatment-effect bounds. Imbens and Manski (2004) distinguish confidence regions for an identified set from confidence intervals for a partially identified parameter.
- Lee (2009) derives sharp treatment-effect bounds under sample selection using random assignment, monotonicity, and outcome-tail trimming.
- Khan, Saveski, and Ugander (2024) obtain sharp smoothness-based bounds for off-policy evaluation beyond overlap.
- Cole and Hernán (2008) discuss practical inverse-probability weight construction and truncation. Petersen et al. (2012) organize diagnostics and responses to positivity violations, and D'Amour et al. (2021) analyze overlap in high-dimensional covariate spaces.
- Arbour et al. (2026) recast extrapolation as a regularized bias–bias–variance tradeoff and propose sensitivity analysis for dependence on parametric modeling assumptions.

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